

VARAD P

vparadkarj@gmail.com • (980) 371 6640 • [linkedin.com/in/vparadkar](https://www.linkedin.com/in/vparadkar) • Alpharetta, GA

SUMMARY

Software engineer with 5+ years of experience building real-time, data-intensive web applications and cloud-connected systems, from React dashboards and RESTful APIs to streaming pipelines and CI/CD. Skilled in React, JavaScript, Python, Java, AWS, and Azure, with a track record of optimizing performance for high-volume data and leading cross-functional teams on UI architecture, code quality, and engineering standards.

EDUCATION

Master of Science, Computer Science | **University of North Carolina at Charlotte, NC** **Aug 2024 - May 2026**

TECHNICAL SKILLS

- **Frontend:** React, JavaScript, GraphQL, TypeScript, Component-Based Architecture, Responsive UI, Performance Optimization
- **Programming:** Python, Java, SQL (PostgreSQL, MySQL), C#, C++, Shell/CLI
- **APIs & Architecture:** RESTful APIs, API Integration, Service-Oriented Architecture, Distributed Systems, Schema Validation
- **Real-Time & Data Systems:** Streaming & Batch Processing, Event-Driven Pipelines, Apache Spark, PySpark, Airflow, Delta Lake
- **Dashboards & Visualization:** React Dashboards, Streamlit, Power BI, Real-Time Data Visualization
- **Cloud & DevOps:** AWS (S3, Glue, EMR), Azure, Microsoft Fabric, Docker, Terraform, CloudFormation, GitHub Actions, Azure DevOps
- **Methodologies:** Monitoring & Logging, Root Cause Analysis, SLA Management, Technical Leadership, Code Review, Agile/Scrum

PROFESSIONAL EXPERIENCE

Research Data Scientist, Urban Institute, Charlotte, NC **May 2025 - May 2026**

- Built React-based KPI dashboards integrated with RESTful APIs, giving non-technical stakeholders clear, real-time views of operational metrics.
- Developed backend APIs and cloud-native pipelines processing 100GB+ of daily data using Python and PySpark, improving validation throughput by 70% and powering the dashboards' data layer.
- Implemented automated monitoring, schema validation, and logging frameworks, significantly reducing data discrepancies surfaced to end users.
- Diagnosed application and pipeline performance bottlenecks through root-cause analysis, working with analysts and business stakeholders to improve SLA compliance.

Distributed Systems Engineer, WiNS Lab, Charlotte, NC **Dec 2024 - May 2025**

- Built distributed real-time streaming and batch pipelines using Python and Apache Spark, supporting high-frequency data workflows and improving model accuracy by 15%.
- Automated deployment and testing through CI/CD pipelines and Docker-based environments, reducing release cycle times by 40% while improving release consistency.
- Owned observability by developing monitoring, logging, and validation frameworks that accelerated issue resolution across production-scale systems.

Data Engineer, Montezuma AI, Pune, India **Aug 2023 - May 2024**

- Engineered cloud-native ingestion and transformation systems processing 10TB+ of enterprise data daily using PySpark, Delta Lake, and Airflow.
- Optimized Spark execution plans and data models, reducing infrastructure costs by 25% while improving query performance for business-critical operational workloads.
- Implemented incremental processing, schema evolution, and partition optimization, improving data freshness and reliability for downstream consumers.

Lead Software Engineer, Jnana Prabodhini, Pune, India **Jan 2019 - Dec 2022**

- Led cross-functional teams building scalable web and mobile applications for 50K+ users, developing React front ends backed by RESTful APIs and increasing feature delivery velocity by 40%.
- Mentored engineers on UI architecture, code quality, and documentation, and established reusable component and service architecture standards across the platform.
- Built CI/CD pipelines, automated testing, and release management processes that reduced production defects and increased deployment reliability.

PROJECTS

Honeywell HVAC GraphRAG Pipeline: Python | Neo4j | Cypher | LangChain | Groq API | Streamlit **Jan 2026 - May 2026**

- Built a hybrid GraphRAG system with a Streamlit interface that combines Neo4j graph traversal with vector search to answer multi-hop product compatibility queries.
- Engineered parallel retrieval branches that execute in under 200ms, keeping end-to-end response latency to 1-2 seconds while raising retrieval accuracy from 76% to 100%.
- Enforced a strict response schema that eliminated hallucinations and returned actionable fallback suggestions when no match was found.

ForgePilot (Autonomous Coding Agent): Python | Groq API | CLI Framework | Agentic Workflows | Automation **Feb 2026 - Mar 2026**

- Built an autonomous AI coding agent that automates multi-file codebase analysis and refactoring, reducing manual debugging time by an estimated 30%.
- Designed stateful agent workflows with secure file system operations and optimized context-window management for large codebases.
- Implemented fault-tolerant execution with autonomous error recovery for reliable, deterministic outputs.